Oscillatory Motion

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Oscillations and Mechanical Waves

Periodic motion is the repeating motion of an object in which it continues to return to a given position after a fixed time interval.

The repetitive movements are called ***oscillations.***

A special case of periodic motion called ***simple harmonic motion*** will be the focus.

- Simple harmonic motion also forms the basis for understanding mechanical waves.

Oscillations and waves also explain many other phenomena quantity.

- Oscillations of bridges and skyscrapers
- Radio and television
- Understanding atomic theory

Periodic Motion

$x = A\sin(at + b)$ as x is periodic function

$x(t + T) = x(t)$

$A\sin(at + aT + b) = A\sin(at + b)$

$aT = 2\pi$

$a = \frac{2\pi}{T}$ = angular frequency = $2\pi f = \omega$

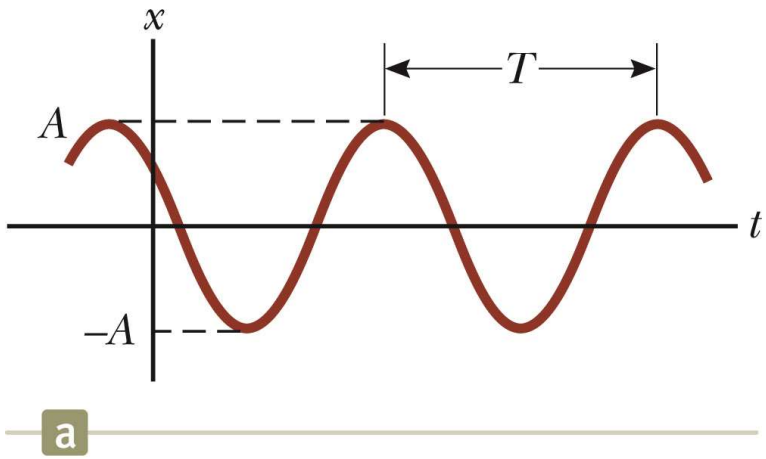
b = *phase const.* = *initial phase*

Simple Harmonic Motion – Graphical Representation

A solution is $x(t) = A \cos(w\,t + \alpha)$

A, w, α are all constants

A cosine curve can be used to give physical significance to these constants.



Simple Harmonic Motion

A and α are determined uniquely by the position and velocity of the particle at $t = 0$.

- If the particle is at $x = A$ at $t = 0$, then $\alpha = 0$

The **phase** of the motion is the quantity $(\omega t + \alpha)$.

$x(t)$ is periodic and its value is the same each time ωt increases by 2π radians.

Period & Frequency

The ***period***, *T*, of the motion is the time interval required for the particle to go through one full cycle of its motion.

$$T = \frac{2\pi}{\omega}$$

The ***frequency*** represents the number of oscillations that the particle undergoes per unit time interval.

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

Units are cycles per second = hertz (Hz).

Motion Equations for Simple Harmonic Motion

$$x(t) = A\cos(\omega t + \phi)$$

$$v = \frac{dx}{dt} = -\omega A\sin(\omega t + \phi)$$

$$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = -\omega^2 A\cos(\omega t + \phi).$$

$$a = -\omega^2 x$$

$$v_{max} = A\omega$$

$$a_{max} = A\omega^2$$

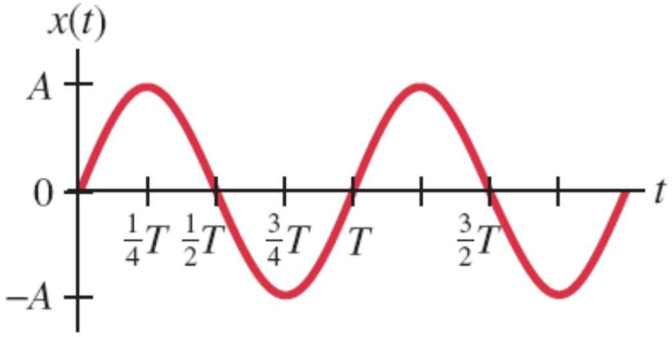
- Simple harmonic motion is one-dimensional and so directions can be denoted by + or - sign.
- Remember, simple harmonic motion is **not** uniformly accelerated motion.

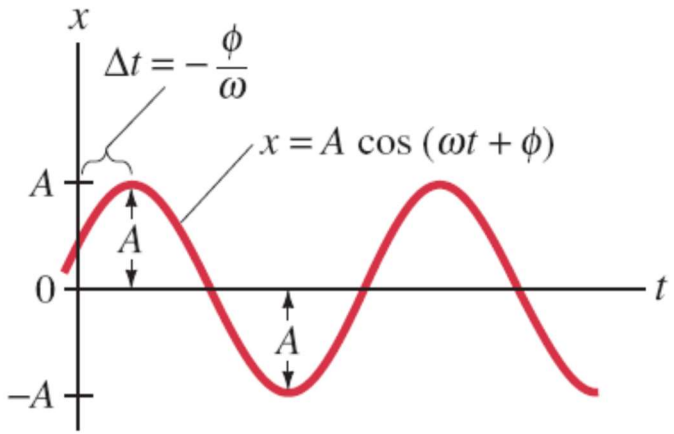
Simple Harmonic Motion

The velocity can be found by differentiating the displacement:

$$v = \frac{dx}{dt} = \frac{d}{dt}[A\cos(\omega t + \phi)] = -\omega A \sin(\omega t + \phi).$$

These figures illustrate the effect of φ:



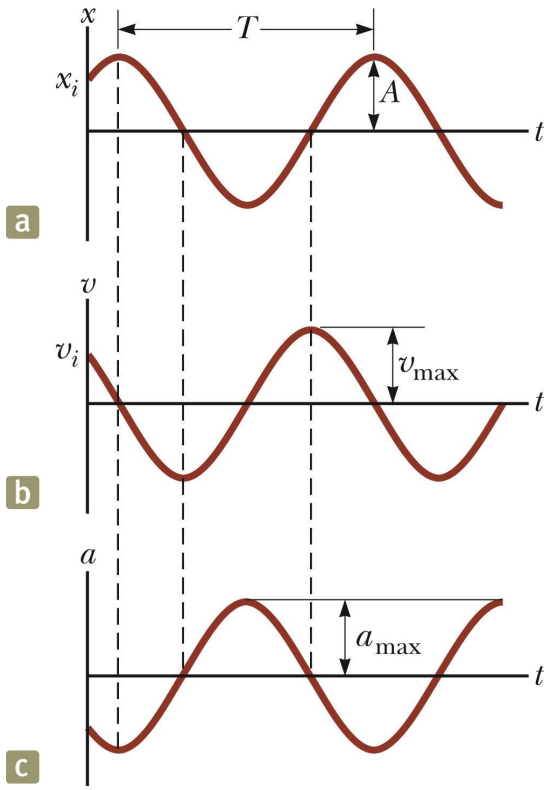


Graphs of Simple Harmonic Motion

The graphs show:

- (a) displacement as a function of time
- (b) velocity as a function of time
- (c) acceleration as a function of time

The velocity is 90° out of phase with the displacement and the acceleration is 180° out of phase with the displacement.



Energy of the SHM Oscillator

The kinetic energy can be found by

- $K = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \phi)$
  - Assume a massless spring, so the mass is the mass of the block.

The elastic potential energy can be found by

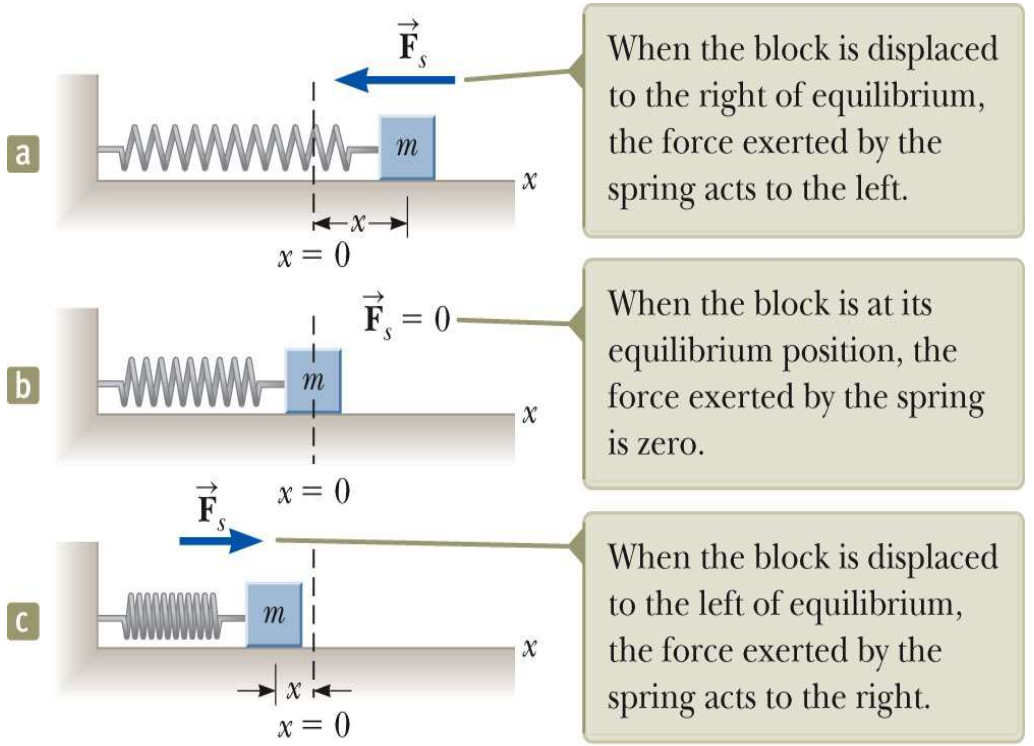
- $U = -\int F dx = -\int ma dx = \int m\omega^2 x dx = 1/2 m\omega^2 x^2$
- $U = \frac{1}{2} m\omega^2 A^2 \cos^2(\omega t + \phi)$

The total energy is $E = K + U = \frac{1}{2} m\omega^2 A^2$

This tells us the total energy is constant.

Motion of a Spring-Mass System

- A block of mass *m* is attached to a spring, the block is free to move on a frictionless horizontal surface.
- When the spring is neither stretched nor compressed, the block is at the ***equilibrium position.***
  - $x = 0$
- Such a system will oscillate back and forth if disturbed from its equilibrium position.



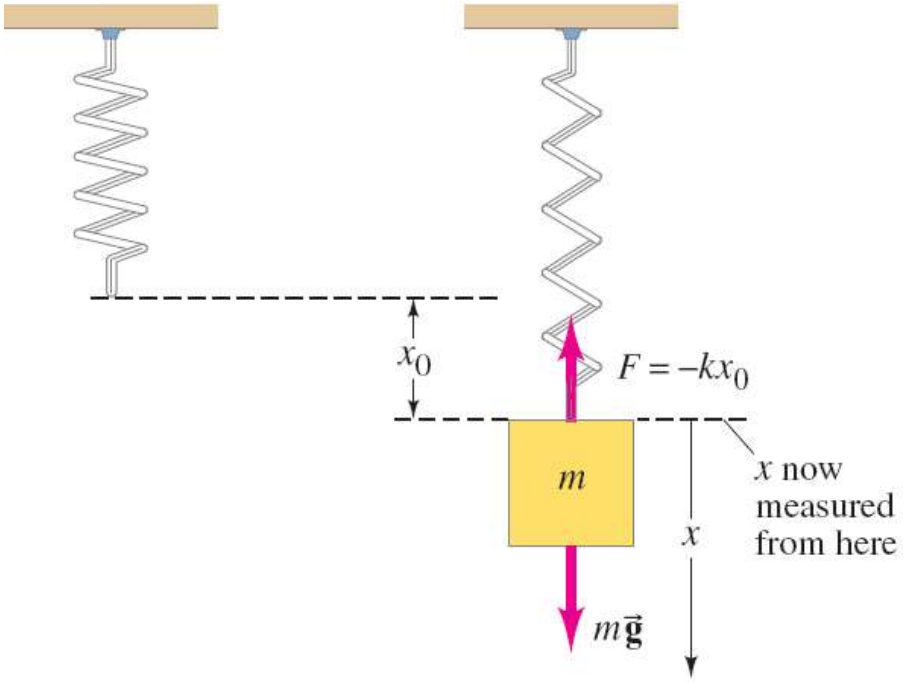
Hooke's Law

Hooke's Law states $F = -kx.$

- F_s is the restoring force.
  - It is always directed toward the equilibrium position.
  - Therefore, it is always opposite the displacement from equilibrium.
- k is the force (spring) constant.
- x is the displacement.

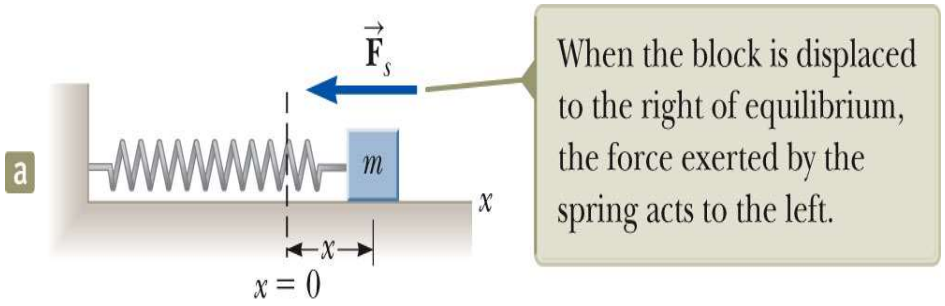
Oscillations of a Spring

If the spring is hung vertically, the only change is in the equilibrium position, which is at the point where the spring force equals the gravitational force.

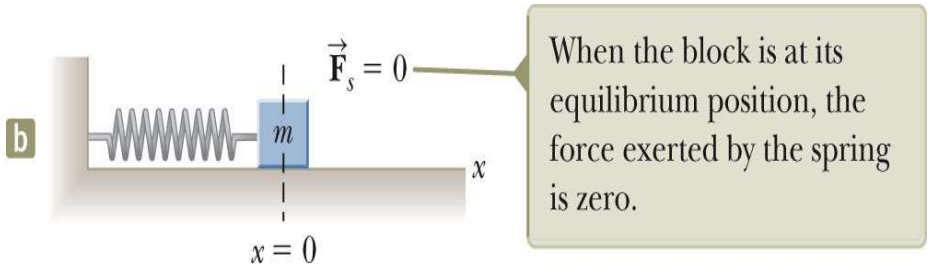


Restoring Force & Spring Mass System

- In a, the block is displaced to the right of $x = 0$.
  - The position is positive.
  - The restoring force is directed to the left.



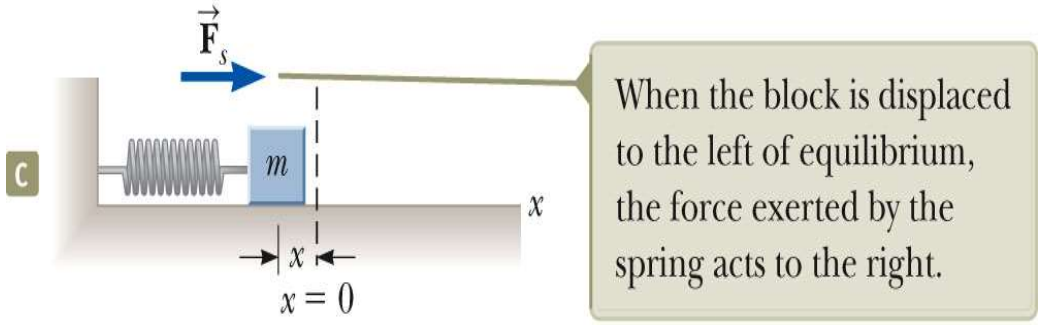
- In b, the block is at the equilibrium position.
  - $x = 0$
  - The spring is neither stretched nor compressed.
  - The force is 0.



Restoring Force

The block is displaced to the left of $x = 0$.

- The position is negative.
- The restoring force is directed to the right.



Acceleration

When the block is displaced from the equilibrium point and released, it is a particle under a net force and therefore has an acceleration.

The force described by Hooke's Law is the net force in Newton's Second Law.

$$-kx = ma_x$$

The direction of the acceleration is opposite the direction of the displacement from equilibrium.

An object moves with simple harmonic motion whenever its acceleration is proportional to its position and is oppositely directed to the displacement from equilibrium.

$$a = -\omega^2 x$$

Spring Mass System, SHM

Model the block as a particle.

- The representation will be **particle in simple harmonic motion model.**

Choose *x* as the axis along which the oscillation occurs.

Acceleration $a = \frac{d^2x}{dt^2} = -\frac{k}{m}x$

Then $a = -\omega^2 x$, *where*

It satisfies SHM Equation

$$\omega = \sqrt{\frac{k}{m}}$$

Maximum Values of *v* and *a*

- Because the sine and cosine functions oscillate between ±1, we can easily find the maximum values of velocity and acceleration for an object in SHM.

$$v_{max} = \omega A = \sqrt{\frac{k}{m}}\, A$$

$$a_{max} = \omega^2 A = \frac{k}{m} A$$

Energy in the Simple Harmonic Oscillator

We already know that the potential energy of a spring is given by:

$$U = -\int F\,dx = \tfrac{1}{2}kx^2.$$

The total mechanical energy is then:

$$E = \tfrac{1}{2}mv^2 + \tfrac{1}{2}kx^2.$$

The total mechanical energy will be conserved, as we are assuming the system is frictionless.

Energy in the Simple Harmonic Oscillator

The total energy is, therefore, $\frac{1}{2}kA^2$.

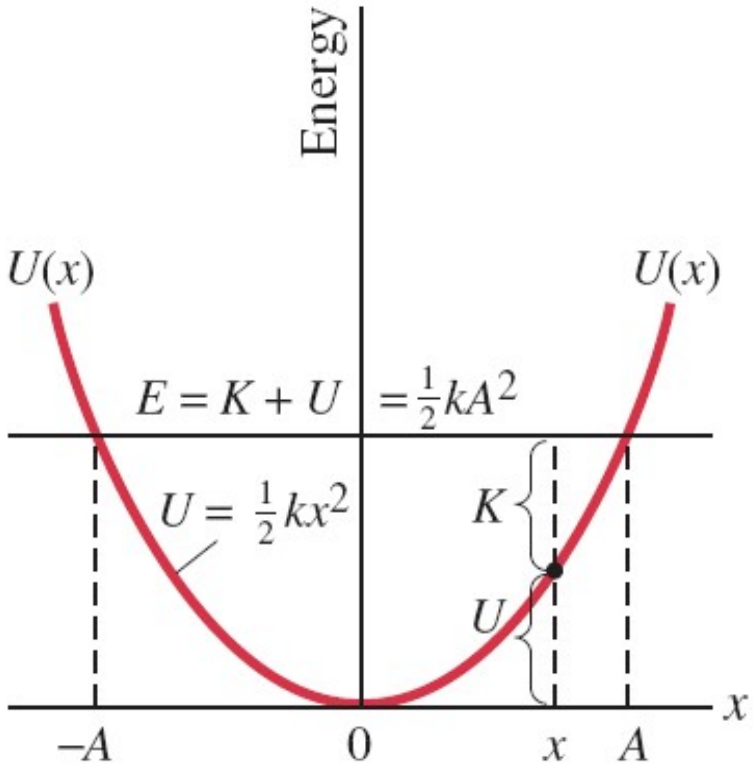
And we can write: $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$.

This can be solved for the velocity as a function of position:

$$v = \pm v_{\max}\sqrt{1 - \frac{x^2}{A^2}}, \qquad \text{where} \qquad v_{\max}^2 = (k/m)A^2.$$

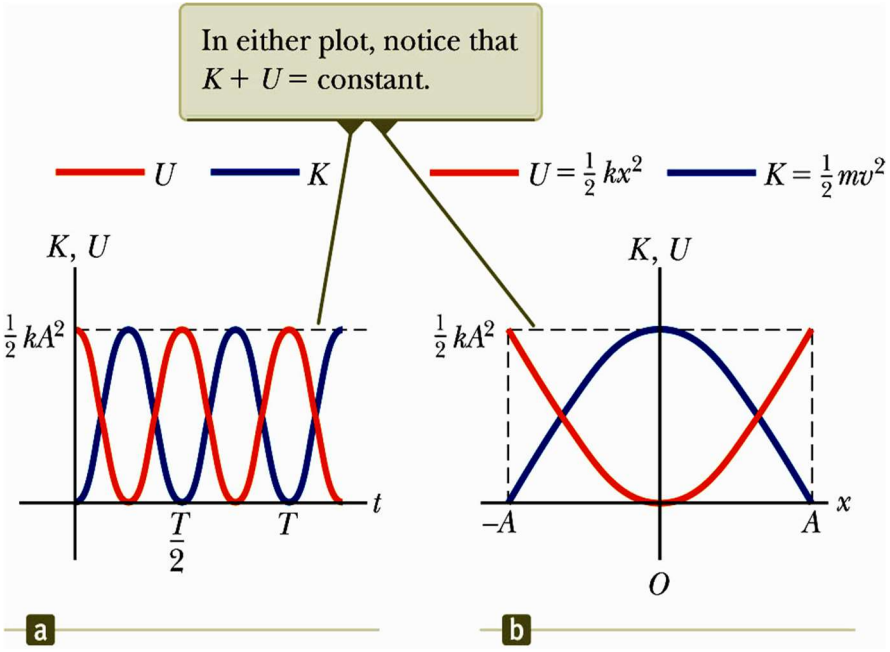
Energy in the Simple Harmonic Oscillator

This graph shows the potential energy function of a spring. The total energy is constant.

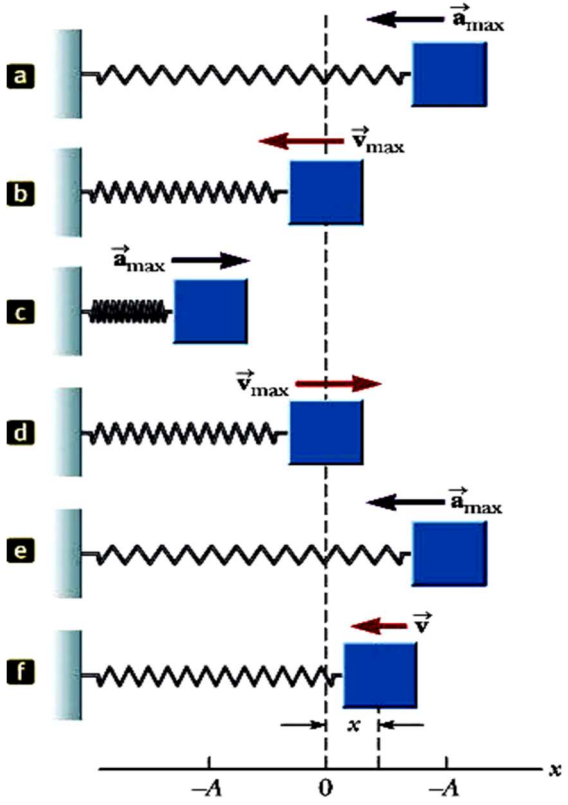


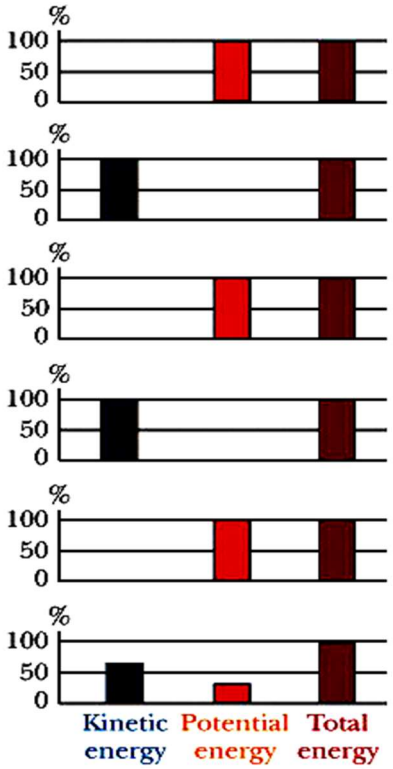
Energy of the SHM Oscillator

- The total mechanical energy is constant.
  - At all times, the total energy is ½ k A^2
- The total mechanical energy is proportional to the square of the amplitude.
- Energy is continuously being transferred between potential energy stored in the spring and the kinetic energy of the block.
- In the diagram, *phase const.* = 0



Energy in Simple Harmonic Motion



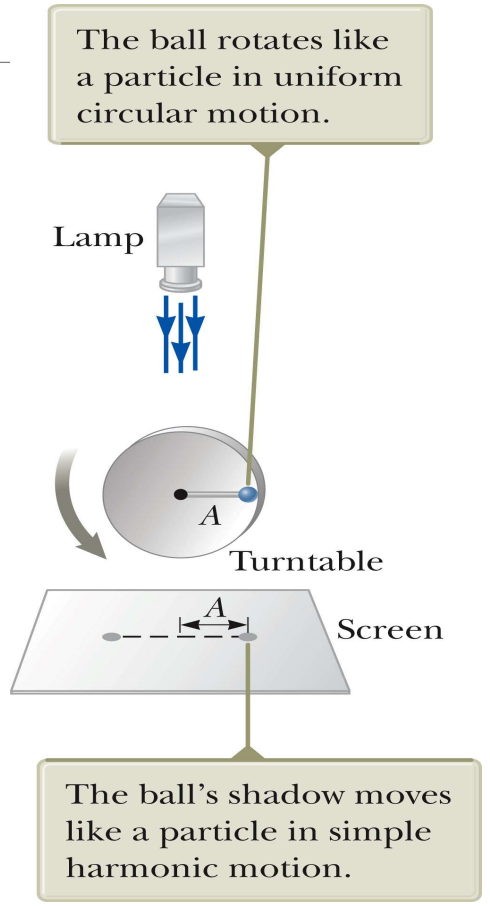


t	x	v	a	K	U
0	A	0	$-\omega^2 A$	0	$\frac{1}{2}kA^2$
$\frac{T}{4}$	0	$-\omega A$	0	$\frac{1}{2}kA^2$	0
$\frac{T}{2}$	$-A$	0	$\omega^2 A$	0	$\frac{1}{2}kA^2$
$\frac{3T}{4}$	0	ωA	0	$\frac{1}{2}kA^2$	0
T	A	0	$-\omega^2 A$	0	$\frac{1}{2}kA^2$
t	x	v	$-\omega^2 x$	$\frac{1}{2}mv^2$	$\frac{1}{2}kx^2$

SHM and Circular Motion

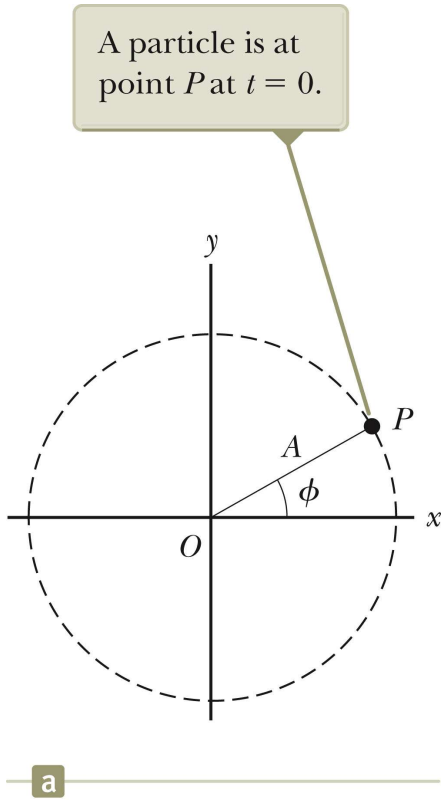
- This is an overhead view of an experimental arrangement that shows the relationship between SHM and circular motion.

- As the turntable rotates with constant angular speed, the ball's shadow moves back and forth in simple harmonic motion.



SHM and Circular Motion

- The circle is called a ***reference circle.***
- For comparing simple harmonic motion and uniform circular motion.
- Take *P* at *t* = 0 as the reference position.
- Line *OP* makes an angle ϕ with the *x* axis at *t* = 0.



SHM and Circular Motion

The particle moves along the circle with constant angular velocity ω.

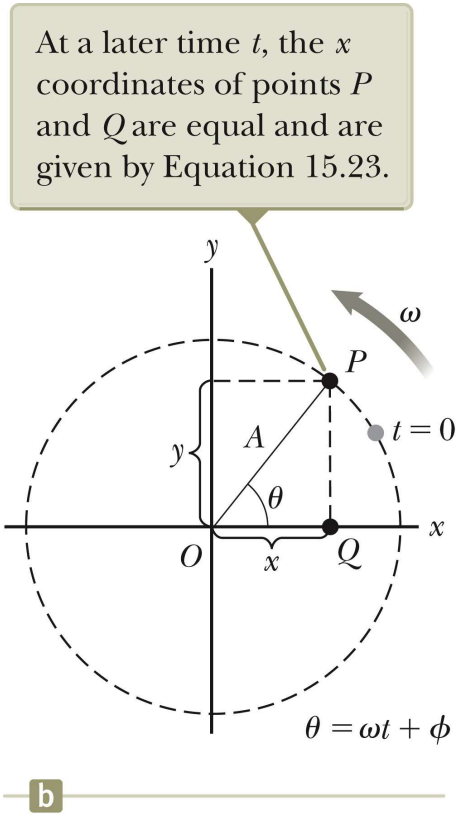
OP makes an angle θ with the *x* axis.

At some time, the angle between *OP* and the *x* axis will be $\theta = \omega t + \phi$.

The points *P* and *Q* always have the same *x* coordinate.

$x(t) = A\cos(\omega t + \phi)$

This shows that point *Q* moves with simple harmonic motion along the *x* axis.

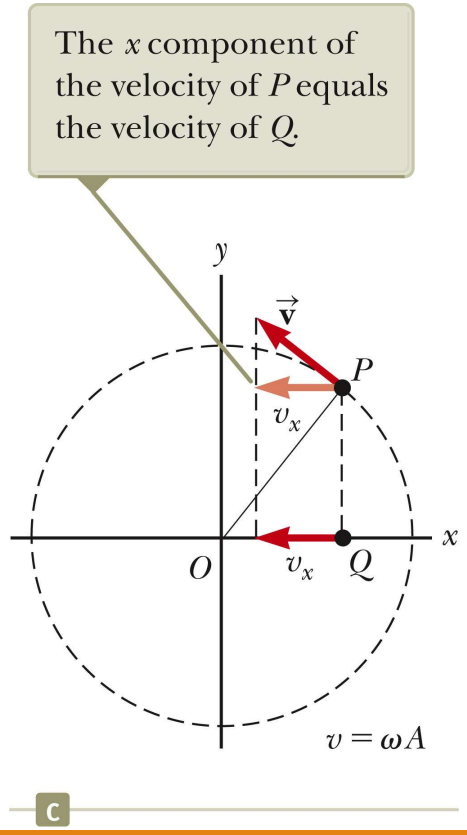


SHM and Circular Motion

- The angular speed of P is the same as the angular frequency of simple harmonic motion along the x axis.
- Point Q has the same velocity as the x component of point P.
- The x-component of the velocity is

$$v = -\omega A \sin(\omega t + \phi)$$

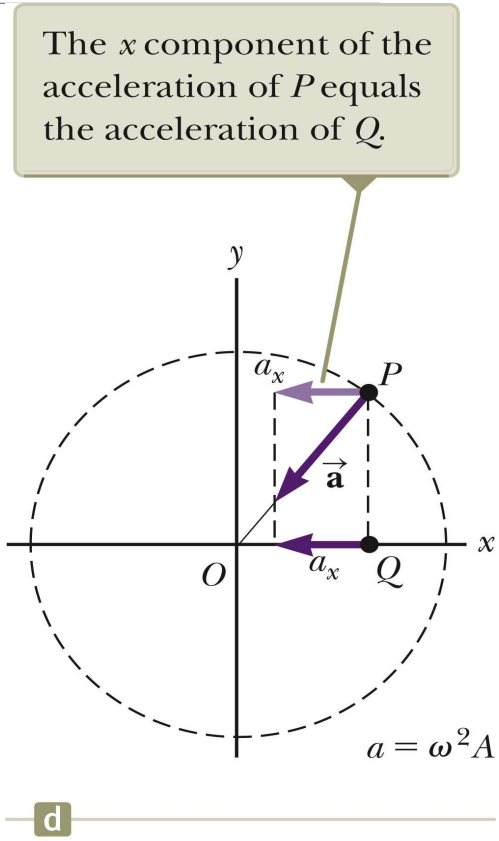
$$v = v_M \sqrt{1 - \frac{x^2}{A^2}}.$$



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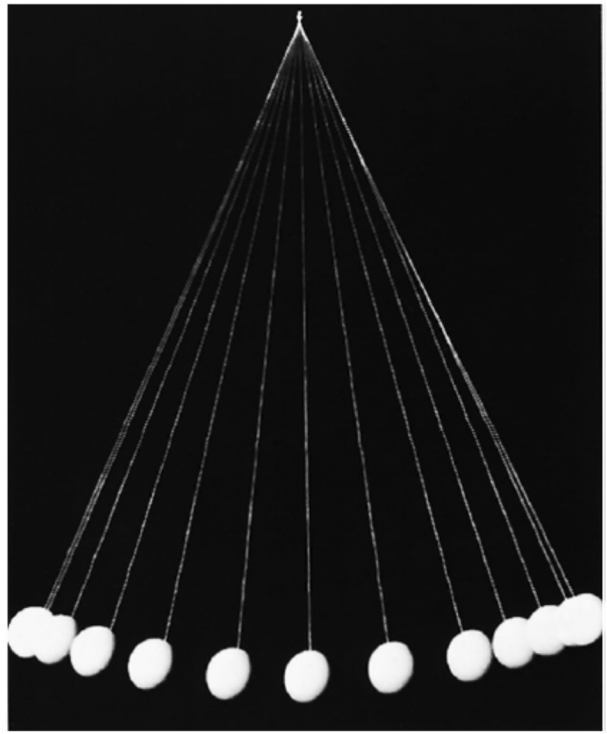
SHM and Circular Motion

- The acceleration of point *P* on the reference circle is directed radially inward.
- *P* 's acceleration is $a = \omega^2 A$
- The *x* component is $-\omega^2 A \cos(\omega t + \phi)$
- This is also the acceleration of point *Q* along the *x* axis.



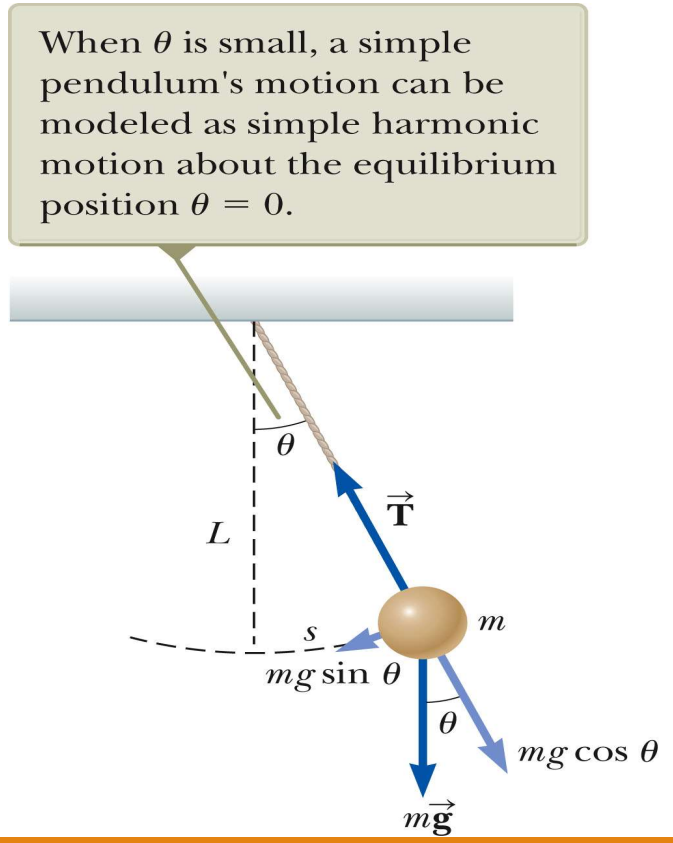
Simple Pendulum

- A simple pendulum also exhibits periodic motion.
- It consists of a particle-like bob of mass *m* suspended by a light string of length *L.*
- The motion occurs in the vertical plane and is driven by gravitational force.
- The motion is very close to that of the SHM oscillator. If the angle is $<10^{o}$



Simple Pendulum

- The forces acting on the bob are the tension and the weight.
  - $\vec{\mathbf{T}}$ is the force exerted on the bob by the string.
  - $m\vec{\mathbf{g}}$ is the gravitational force.
- The tangential component of the gravitational force is a restoring force.



Simple Pendulum

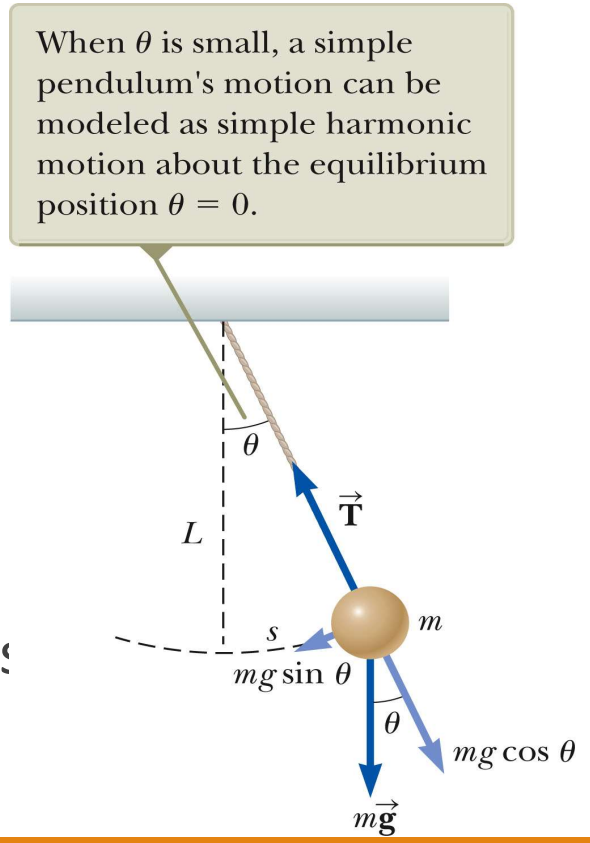
In the tangential direction,

$$F_t = ma_t \rightarrow -mg\sin\theta = m\frac{d^2s}{dt^2}$$

The length, *L*, of the pendulum is constant, and for small values of θ.

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin\theta = -\frac{g}{L}\theta$$

This confirms the mathematical form of the motion is the same as for SHM.



Simple Pendulum

The function θ can be written as $\theta = \theta_{max} \cos(\omega t + \phi)$.

The angular frequency is $\omega = \sqrt{\frac{g}{L}}$ The period is $T = 2\pi\sqrt{\frac{L}{g}}$,

- The period and frequency of a simple pendulum depend only on the length of the string and the acceleration due to gravity.
- The period is independent of the mass.
- All simple pendula that are of equal length and are at the same location oscillate with the same period.

Example 1:

Car springs

Determine the period and frequency of a car whose mass is 1400 kg and whose shock absorbers have a spring constant of 6.5×10^4 N/m after hitting a bump. Assume the shock absorbers are poor, so the car really oscillates up and down.

T = 0.92 s and f = 1.09 Hz.

Example 2:

A vibrating floor.

A large motor in a factory causes the floor to vibrate at a frequency of 10 Hz. The amplitude of the floor's motion near the motor is about 3.0 mm. Estimate the maximum acceleration of the floor near the motor.

Solution: Assuming the motion is simple harmonic, the maximum acceleration is $\omega^2 A$, where $\omega = 2\pi f$. This gives $a = 12\ m/s^2$.

Example 4:

The cone of a loudspeaker oscillates in SHM at a frequency of 262 Hz ("middle C"). The amplitude at the center of the cone is A = 1.5 x 10^{-4} m, and at t = 0, $x = A$. (a) What equation describes the motion of the center of the cone? (b) What are the velocity and acceleration as a function of time? (c) What is the position of the cone at t = 1.00 ms ?



Solution: a. The angular frequency is 1650 rad/s, so x = (1.5 x 10^{-4}m)cos(1650t).
b. The maximum velocity is 0.25 m/s, so v = -(0.25 m/s)sin(1650t).
The maximum acceleration is 410 m/s^2, so a = -(410 m/s^2)cos(1650t).
c. At t = 1 ms, x = -1.2 x 10^{-5} m.

Simple Harmonic Motion, Examples

Example: Energy calculations.

For the simple harmonic oscillation (where k = 19.6 N/m, A = 0.100 m, x = -(0.100 m) cos 8.08t, and v = (0.808 m/s) sin 8.08t), determine (a) the total energy, (b) the kinetic and potential energies as a function of time, (c) the velocity when the mass is 0.050 m from equilibrium, (d) the kinetic and potential energies at half amplitude ($x = \pm A/2$).

Solution: a. $E = \frac{1}{2} kA^2 = 9.80 \times 10^{-2}$ J.
b. $U = \frac{1}{2} kx^2 = (9.80 \times 10^{-2}\ \text{J}) \cos^2 8.08t$, $K = \frac{1}{2} mv^2 = (9.80 \times 10^{-2}\ \text{J}) \sin^2 8.08t$.
c. v = 0.70 m/s
d. $U = 2.5 \times 10^{-2}$ J, $K = E - U = 7.3 \times 10^{-2}$ J.

Example 5:

A spring stretches 0.15 m when a 0.3 kg mass is gently attached to it. The spring is then set up horizontally with the 0.3kg mass resting on a frictionless table. The mass is pushed so that the spring is compressed 0.100 m from the equilibrium point, and released from rest.
Determine: (a) the spring stiffness constant k and angular frequency ω; (b) the amplitude of the horizontal oscillation A; (c) the magnitude of the maximum velocity $v_{\max}$; (d) the magnitude of the maximum acceleration $a_{\max}$ of the mass; (e) the period T and frequency f; (f) the displacement x as a function of time; and (g) the velocity at t = 0.150 s.

Solution

A. The spring constant is 19.6 N/m, so $\omega = 8.08/s^{-1}$.
b. Since the spring is released from rest, A = 0.100 m.
c. The maximum velocity is 0.808 m/s
d. The maximum acceleration occurs when the displacement is maximum, and equals 6.53 m/s^2.
e. The period is 0.777 s, and the frequency is 1.29 Hz.
f. At $t = 0$, x = -A; this is a negative cosine.

$x = (0.100\ m)\cos(8.08t - \pi)$.

g. V = dx/dt; at 0.150 s it is 0.756 m/s.

Example 6:

Example: Measuring g.

A geologist uses a simple pendulum that has a length of 37.10 cm and a frequency of 0.8190 Hz at a particular location on the Earth. What is the acceleration of gravity at this location?

$g = 9.824\ m/s^2$.